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```
MariaDB [hotelSystem]> create database project_schema;  
Query OK, 1 row affected (0.002 sec)
```

```
MariaDB [hotelSystem]> use project_schema;  
Database changed
```

```
MariaDB [project_schema]> create table employee(  
    -> empNo varchar(3) not null,  
    -> fName varchar(50),  
    -> lName varchar(50),  
    -> address varchar (200),  
    -> DOB date,  
    -> sex char(1) not null check (sex in ('M','F')),  
    -> position varchar(30) check (position in ('Manager', 'Team Leader',  
'Analyst', 'Software Developer')),  
    -> Primary key (empNo));  
Query OK, 0 rows affected (0.119 sec)
```

```
MariaDB [project_schema]> create table Department(  
    -> deptNo varchar(3) not null,  
    -> deptName varchar(50),  
    -> mgrEmpNo varchar(3) not null,  
    -> Primary key (deptNo),  
    -> Foreign key (mgrEmpNo) references Employee(empNo));  
Query OK, 0 rows affected (0.124 sec)
```

```
MariaDB [project_schema]> alter table employee add column depNo varchar(3);  
Query OK, 0 rows affected (0.262 sec)  
Records: 0 Duplicates: 0 Warnings: 0
```

```
MariaDB [project_schema]> alter table employee  
    -> add constraint fk_depNo  
    -> Foreign key (depNo) references employee(empNo);  
Query OK, 0 rows affected (0.240 sec)  
Records: 0 Duplicates: 0 Warnings: 0
```

```
MariaDB [project_schema]> create table Project(  
    -> projNo varchar(3) not null,  
    -> projName varchar(50),  
    -> deptNo varchar(3),  
    -> Primary key(projNo),  
    -> Foreign key (deptNo) references Department(deptNo));  
Query OK, 0 rows affected (0.120 sec)
```

```
MariaDB [project_schema]> create table WorksOn(  
    -> empNo varchar(3) Not null,  
    -> projNo varchar(3),  
    -> dateWorked date,  
    -> hoursWorked int(2) check(hoursWorked between 0 and 40),  
    -> Primary key (empNo, projNo, dateWorked),  
    -> Foreign key(empNo) references Employee(empNo),  
    -> Foreign key(projNo) references Project(projNo));  
Query OK, 0 rows affected (0.119 sec)
```

```
MariaDB [project_schema]> ALTER TABLE employee DROP FOREIGN KEY fk_depNo;  
Query OK, 0 rows affected (0.269 sec)  
Records: 0 Duplicates: 0 Warnings: 0
```

```
MariaDB [project_schema]> ALTER TABLE employee  
    -> ADD CONSTRAINT fk_depNo  
    -> FOREIGN KEY (depNo) REFERENCES Department(deptNo);
```

Query OK, 0 rows affected (0.234 sec)
Records: 0 Duplicates: 0 Warnings: 0

MariaDB [project_schema]> SET FOREIGN_KEY_CHECKS = 0;
Query OK, 0 rows affected (0.002 sec)

MariaDB [project_schema]> INSERT INTO Department (deptNo, deptName, mgrEmpNo)
VALUES

- > ('D1', 'Human Resources', 'E1'),
- > ('D2', 'Research & Development', 'E2'),
- > ('D3', 'Information Technology', 'E3'),
- > ('D4', 'Marketing & Sales', 'E4'),
- > ('D5', 'Finance & Accounting', 'E5'),
- > ('D6', 'Customer Operations', 'E6'),
- > ('D7', 'Data Analytics', 'E18');

Query OK, 7 rows affected (0.002 sec)
Records: 7 Duplicates: 0 Warnings: 0

MariaDB [project_schema]> INSERT INTO employee (empNo, fName, lName, address,
DOB, sex, position, depNo) VALUES

- > ('E1', 'John', 'Doe', '123 Elm St, Accra', '1980-05-15', 'M', 'Manager', 'D1'),
- > ('E2', 'Jane', 'Smith', '456 Oak St, Kumasi', '1985-08-22', 'F', 'Manager', 'D2'),
- > ('E3', 'Robert', 'Johnson', '789 Pine St, Tamale', '1978-11-30', 'M', 'Manager', 'D3'),
- > ('E4', 'Emily', 'Davis', '321 Maple Dr, Takoradi', '1990-03-10', 'F', 'Team Leader', 'D4'),
- > ('E5', 'Michael', 'Brown', '654 Cedar Ln, Tema', '1992-07-19', 'M', 'Team Leader', 'D5'),
- > ('E6', 'Sarah', 'Wilson', '987 Birch Ave, Cape Coast', '1988-12-05', 'F', 'Team Leader', 'D6'),
- > ('E7', 'David', 'Taylor', '147 Palm St, Accra', '1995-01-25', 'M', 'Analyst', 'D1'),
- > ('E8', 'Amanda', 'Thomas', '258 Rose Rd, Kumasi', '1993-09-14', 'F', 'Analyst', 'D1'),
- > ('E9', 'James', 'Anderson', '369 Lily Ln, Sunyani', '1994-04-18', 'M', 'Analyst', 'D2'),
- > ('E10', 'Jessica', 'Jackson', '159 Sunflower Dr, Koforidua', '1991-06-29', 'F', 'Software Developer', 'D2'),
- > ('E11', 'Daniel', 'White', '260 Orchid St, Ho', '1996-10-12', 'M', 'Software Developer', 'D2'),
- > ('E12', 'Megan', 'Harris', '361 Daisy Rd, Bolgatanga', '1997-02-03', 'F', 'Software Developer', 'D3'),
- > ('E13', 'Kevin', 'Martin', '462 Jasmine Ct, Wa', '1995-11-21', 'M', 'Software Developer', 'D3'),
- > ('E14', 'Rachel', 'Clark', '563 Tulip Ave, Techiman', '1993-08-08', 'F', 'Software Developer', 'D3'),
- > ('E15', 'William', 'Lewis', '664 Marigold Blvd, Tema', '1994-12-12', 'M', 'Software Developer', 'D3'),
- > ('E16', 'Oliver', 'Green', '775 Lavender Ln, Kasoa', '1995-04-11', 'M', 'Software Developer', 'D4'),
- > ('E17', 'Laura', 'Baker', '886 Mint St, Ashaiman', '1996-07-22', 'F', 'Analyst', 'D5'),
- > ('E18', 'Simon', 'Ward', '997 Sage Rd, Obuasi', '1992-10-05', 'M', 'Team Leader', 'D7');

Query OK, 18 rows affected (0.002 sec)
Records: 18 Duplicates: 0 Warnings: 0

MariaDB [project_schema]> INSERT INTO Project (projNo, projName, deptNo) VALUES

- > ('P1', 'Talent Acquisition System', 'D1'),
- > ('P2', 'Employee Wellness Initiative', 'D1'),
- > ('P3', 'Next-Gen Core Engine', 'D2'),
- > ('P4', 'AI Predictive Modeling', 'D2'),

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-> ('P5', 'Cloud Infrastructure Migration', 'D3'),
-> ('P6', 'Database Security Hardening', 'D3'),
-> ('P7', 'Global Brand Campaign', 'D4'),
-> ('P8', 'Digital Ads Optimization', 'D4'),
-> ('P9', 'Automated Billing Portal', 'D5'),
-> ('P10', 'Risk Assessment Engine', 'D5'),
-> ('P11', 'CRM Gateway Upgrade', 'D6'),
-> ('P12', 'Support Ticket Bot', 'D6'),
-> ('P13', 'BI Metrics Dashboard', 'D7'),
-> ('P14', 'Data Pipeline Processing', 'D7'),
-> ('P15', 'Enterprise Firewall Upgrade', 'D3');

```

Query OK, 15 rows affected (0.002 sec)
Records: 15 Duplicates: 0 Warnings: 0

MariaDB [project_schema]> INSERT INTO WorksOn (empNo, projNo, dateWorked, hoursWorked) VALUES

```

-> ('E4', 'P1', '2026-01-10', 35),
-> ('E4', 'P2', '2026-01-17', 5),
-> ('E7', 'P1', '2026-01-10', 40),
-> ('E8', 'P2', '2026-01-12', 20),
-> ('E8', 'P7', '2026-01-15', 20),
-> ('E5', 'P3', '2026-02-01', 30),
-> ('E5', 'P4', '2026-02-08', 10),
-> ('E9', 'P3', '2026-02-01', 40),
-> ('E10', 'P4', '2026-02-03', 25),
-> ('E10', 'P9', '2026-02-10', 15),
-> ('E11', 'P10', '2026-02-15', 38),
-> ('E6', 'P5', '2026-03-01', 32),
-> ('E6', 'P6', '2026-03-08', 8),
-> ('E12', 'P5', '2026-03-01', 40),
-> ('E13', 'P6', '2026-03-05', 35),
-> ('E13', 'P15', '2026-03-12', 5),
-> ('E14', 'P11', '2026-03-15', 40),
-> ('E15', 'P12', '2026-03-20', 20),
-> ('E16', 'P7', '2026-01-22', 25),
-> ('E16', 'P8', '2026-01-29', 15),
-> ('E17', 'P13', '2026-02-18', 30),
-> ('E17', 'P14', '2026-02-25', 10),
-> ('E18', 'P13', '2026-02-12', 15),
-> ('E14', 'P12', '2026-03-22', 18);

```

Query OK, 24 rows affected (0.002 sec)
Records: 24 Duplicates: 0 Warnings: 0

MariaDB [project_schema]> SET FOREIGN_KEY_CHECKS = 1;
Query OK, 0 rows affected (0.000 sec)

MariaDB [project_schema]> create view femProject as

```

-> select projNo, projName, project.deptNo
-> from project, employee, department
-> where Project.deptNo = Department.deptNo
-> and Employee.empNo = Department.mgrEmpNo
-> and Employee.sex = 'F'
-> order by projNo;

```

Query OK, 0 rows affected (0.098 sec)

MariaDB [project_schema]> select * from femProject

-> ;

projNo	projName	deptNo
P11	CRM Gateway Upgrade	D6
P12	Support Ticket Bot	D6

P3	Next-Gen Core Engine	D2
P4	AI Predictive Modeling	D2
P7	Global Brand Campaign	D4
P8	Digital Ads Optimization	D4

6 rows in set (0.005 sec)

```
MariaDB [project_schema]> create view newProject as
-> select e.empNo, e.fName, e.lName, p.projName, w.hoursWorked
-> from employee e, project p, workson w
-> where e.empNo = w.empNo
-> and p.projNo = w.projNo;
Query OK, 0 rows affected (0.096 sec)
```

```
MariaDB [project_schema]> select * from newproject;
```

empNo	fName	lName	projName	hoursWorked
E10	Jessica	Jackson	AI Predictive Modeling	25
E10	Jessica	Jackson	Automated Billing Portal	15
E11	Daniel	White	Risk Assessment Engine	38
E12	Megan	Harris	Cloud Infrastructure Migration	40
E13	Kevin	Martin	Enterprise Firewall Upgrade	5
E13	Kevin	Martin	Database Security Hardening	35
E14	Rachel	Clark	CRM Gateway Upgrade	40
E14	Rachel	Clark	Support Ticket Bot	18
E15	William	Lewis	Support Ticket Bot	20
E16	Oliver	Green	Global Brand Campaign	25
E16	Oliver	Green	Digital Ads Optimization	15
E17	Laura	Baker	BI Metrics Dashboard	30
E17	Laura	Baker	Data Pipeline Processing	10
E18	Simon	Ward	BI Metrics Dashboard	15
E4	Emily	Davis	Talent Acquisition System	35
E4	Emily	Davis	Employee Wellness Initiative	5
E5	Michael	Brown	Next-Gen Core Engine	30
E5	Michael	Brown	AI Predictive Modeling	10
E6	Sarah	Wilson	Cloud Infrastructure Migration	32
E6	Sarah	Wilson	Database Security Hardening	8
E7	David	Taylor	Talent Acquisition System	40
E8	Amanda	Thomas	Employee Wellness Initiative	20
E8	Amanda	Thomas	Global Brand Campaign	20
E9	James	Anderson	Next-Gen Core Engine	40

24 rows in set (0.001 sec)

```
MariaDB [project_schema]> CREATE VIEW EmpProject(empNo, projNo, totalHours)
-> AS SELECT w.empNo, w.projNo, SUM(hoursWorked)
-> FROM Employee e, Project p, WorksOn w
-> WHERE e.empNo = w.empNo
-> AND p.projNo = w.projNo
-> GROUP BY w.empNo, w.projNo;
Query OK, 0 rows affected (0.143 sec)
```

```
MariaDB [project_schema]> select * from empproject;
```

empNo	projNo	totalHours
E10	P4	25
E10	P9	15
E11	P10	38
E12	P5	40
E13	P15	5

E13	P6	35
E14	P11	40
E14	P12	18
E15	P12	20
E16	P7	25
E16	P8	15
E17	P13	30
E17	P14	10
E18	P13	15
E4	P1	35
E4	P2	5
E5	P3	30
E5	P4	10
E6	P5	32
E6	P6	8
E7	P1	40
E8	P2	20
E8	P7	20
E9	P3	40

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24 rows in set (0.001 sec)

```
MariaDB [project_schema]> select projNo from empproject
-> where projNo = 'SCCS';
Empty set (0.005 sec)
```

```
MariaDB [project_schema]> select count(projNo) from empProject where empNo =
'E1';
```

count(projNo)
0

+-----+

1 row in set (0.004 sec)

```
MariaDB [project_schema]> select empno, totalhours
-> from empproject
-> group by empno;
```

empno	totalhours
E10	25
E11	38
E12	40
E13	5
E14	40
E15	20
E16	25
E17	30
E18	15
E4	35
E5	30
E6	32
E7	40
E8	20
E9	40

+-----+

15 rows in set (0.001 sec)

A materialized view store the view as a temporary table in the database when the

view is first queried.

The base table Part is not needed when the effect of the update does not satisfy the condition in the view.

For instance, $\text{partcost} \leq 1000$ inserted, deleted or updated will not make any changes to the view.

A challenge occurs when $\text{partCost} \geq 1000$ or a modification of partNNo is made due to the distinct key word.